

Local Lipschitz estimate for the amplitude coefficients

Claude, with Daniel Murfet

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Abstract

Completion of Astra #10's rank 5. The amplitude coefficient map $(x, y) \mapsto c(x, y)(s)$ of the §4.2 cutoff integrand, $c_k(s) = e^{\beta s x_{00}} (y * E_{\bar{x}}(\beta s))_k$, is locally Lipschitz in the weighted norm $\|\cdot\|_\rho$:

$$\|c(x, y)(s) - c(x', y')(s)\|_\rho \leq e^{2|\beta s|M_x} (\|y - y'\|_\rho + 2M_y|\beta s|\|x - x'\|_\rho)$$

for $\|x\|_\rho, \|x'\|_\rho \leq M_x, \|y\|_\rho \leq M_y$ (`ampCoeff_lipschitz`), with the pointwise form $|c_k(x, y)(s) - c_k(x', y')(s)|\rho^{|k|} \leq$ the same (`ampCoeff_sub_abs_le`). Zero sorry/axiom.

1 The things formalised

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theorem wnorm_const_mul ( c f ) : wnorm (fun k => c * f k) = |c| * wnorm f
theorem term_le_wnorm ( ) (h : 0 < ) (f) (hf) (k) : |f k| * ^ (k.1 + k.2) wnorm f
theorem dropConst_sub (x x') :
  dropConst (fun k => x k - x' k) = fun k => dropConst x k - dropConst x' k
theorem ampCoeff_sub_decomp ( x x' y y' s ) :
  (fun k => ampCoeff x y k s - ampCoeff x' y' k s)
  = fun k => exp(s x ) * (conv (y - y') E k + conv y' (E - E') k)
    + (exp(s x ) - exp(s x' )) * conv y' E' k
theorem exp_phase_sub_le ... : |exp (t * x (0,0)) - exp (t * x' (0,0))|
  Real.exp (|t| * M) * (|t| * wnorm (fun k => x k - x' k))
theorem ampCoeff_lipschitz ( ) (h : 0 < ) (x x' y y') (hx hx' hy hy') (Mx My)
  (hMx : wnorm x Mx) (hMx' : wnorm x' Mx) (hMy' : wnorm y' My) (s) :
  WSummable (fun k => ampCoeff x y k s - ampCoeff x' y' k s)
  wnorm (fun k => ampCoeff x y k s - ampCoeff x' y' k s)
  Real.exp (2 * | * s| * Mx)
  * (wnorm (fun k => y k - y' k)
    + 2 * My * | * s| * wnorm (fun k => x k - x' k))
theorem ampCoeff_sub_abs_le ... :
  |ampCoeff x y k s - ampCoeff x' y' k s| * ^ (k.1 + k.2) ...

```

2 Mathematics

Write $u = \beta s x_{00}$, $E = E_{\bar{x}}(\beta s)$ with $\bar{x} = \text{dropConst } x$, and primes for the second data set. Then

$$c - c' = e^u (y - y') * E + e^u y' * (E - E') + (e^u - e^{u'}) y' * E'.$$

The norms are controlled by $\|E\|_\rho \leq e^{|t|\|\bar{x}\|_\rho} \leq e^{|t|M_x}$ (unit 118 and $\|\bar{x}\|_\rho \leq \|x\|_\rho$), $\|E - E'\|_\rho \leq |t|e^{|t|M_x}\|\bar{x} - \bar{x}'\|_\rho \leq |t|e^{|t|M_x}\|x - x'\|_\rho$ (unit 142, using $\overline{x - x'} = \bar{x} - \bar{x}'$), $e^u \leq e^{|t|M_x}$ since $|x_{00}| \leq \|x\|_\rho$ (the $k = (0, 0)$ term with weight $\rho^0 = 1$), and $|e^u - e^{u'}| \leq |u - u'|e^{\max(u, u')} \leq e^{|t|M_x}|t|\|x - x'\|_\rho$. Adding the three pieces gives $e^{2|t|M_x}(\|y - y'\|_\rho + 2M_y|t|\|x - x'\|_\rho)$. Only y' needs a norm bound; y enters through $y - y'$ alone.

3 Paper-facing meaning

Astra #10 listed “ $c_s(x, y)$ is locally Lipschitz in the Banach-algebra norm” as the last piece of the coefficient-closure programme: with parameter measurability (units 138–139), the weighted bounds (140) and linearity (141), it makes the map from analytic data (ξ, η) to Taylor-tree coefficients a continuous (indeed locally Lipschitz) map on the ρ -ball, which is what one needs to move expectations and limits past the coefficient extraction in §4.3. The estimate is uniform on $s \in [0, S]$ for each fixed S , with the Gaussian-type growth $e^{2|\beta s|M_x}$ in s that the envelope class of unit 140 absorbs.

4 Lean notes

- The decomposition lemma `ampCoeff_sub_decomp` is proved by `funext`, unfolding `ampCoeff`, rewriting with `conv_sub_left/conv_sub_right` and `ring`; the convolution values are atoms for `ring`, so no further work is needed.
- Introducing `set t := * s` before the main calc and rewriting the goal with `← ht` after `rw [ampCoeff_sub_decomp]` keeps all hypotheses syntactically in terms of `t`, which is what makes the final `add_le_add (mul_le_mul ...)` step type-check against the `haves`.
- `abs_exp_sub_exp_le` (already in `MixedBlockEnvelope`) has the shape $|x - y| e^{\max(x, y)}$; the case split on `le_total` for the maximum is two lines.
- The file compiled on the first attempt, an indication that the convolution-algebra layer of unit 142 is the right abstraction.